

Functional Annotation Report: *pssA* (Q88GQ4, PP_3664) — Phosphatidylserine Synthase of *Pseudomonas putida* KT2440

Summary

The gene *pssA* (UniProt Q88GQ4, ordered locus PP_3664) of *Pseudomonas putida* strain KT2440 encodes **phosphatidylserine synthase (PssA; EC 2.7.8.8)**, a **CDP-diacylglycerol—serine O-phosphatidyltransferase**. Its primary and best-supported function is to catalyze the **committed first step of the phosphatidylethanolamine (PE) biosynthetic pathway**: it transfers the phosphatidyl moiety of **CDP-diacylglycerol (CDP-DG)** onto the hydroxyl of **L-serine**, producing **phosphatidylserine (PS)** and releasing **CMP**. Phosphatidylserine is then decarboxylated by a separately encoded phosphatidylserine decarboxylase (Psd, PP_4908) to yield **phosphatidylethanolamine**, one of the dominant glycerophospholipids of the *P. putida* membrane. This two-enzyme route (CDP-DG → PS → PE) is the canonical bacterial pathway for producing the major zwitterionic membrane lipid.

Mechanistically, PssA is a member of the **phospholipase D (PLD) superfamily** and belongs to the **CDP-alcohol phosphatidyltransferase class-II / PGS1** clade. The 442-residue protein carries the two hallmark **HKD catalytic motifs** (H-x-K-x₄-D) that signature PLD-superfamily transferases; these two half-sites fold together to build a single bilobed active site that operates through a **covalent phosphatidyl–histidine (phosphatidyl-enzyme) intermediate**. It is a **peripheral membrane protein** that is catalytically active only when associated with the cytoplasmic face of the inner membrane and inactive when in the cytosol — a localization-based regulatory switch. These conclusions rest on a coherent, converging evidence base: curated EC/GO/KEGG annotation, InterPro/Pfam domain signatures, direct sequence analysis of Q88GQ4, and strong orthology (**46% amino-acid identity** with conserved catalytic motifs) to the experimentally characterized and crystallized *Escherichia coli* PssA.

Although no dedicated biochemical or genetic study of the *P. putida* KT2440 protein itself was located, the functional assignment transfers with high confidence from well-characterized orthologs (*E. coli*, *Sinorhizobium meliloti*, *Bacillus subtilis*). In these systems, loss of *pssA* abolishes PS synthase activity and PE formation, confirming the enzyme's role as the pathway's

committed entry step. The enzyme additionally shows **acceptor-site promiscuity** — it can use L-homoserine to make phosphatidylhomoserine in *E. coli* — and its expression level directly modulates membrane PE content and stress tolerance in engineered *Pseudomonas* and *E. coli* strains.

Gene/Protein Identity Verification

Before presenting findings, the target identity was verified against the supplied UniProt record; all evidence is consistent.

Attribute	UniProt record (Q88GQ4)	Verification result
Gene symbol	<i>pssA</i>	☐ Matches PssA / phosphatidylserine synthase literature
Enzyme	CDP-diacylglycerol—serine O-phosphatidyltransferase, EC 2.7.8.8	☐ Consistent with GO:0003882 and KEGG K00998
Locus	PP_3664	☐ KEGG ppu:PP_3664; complement(4162133..4163461)
Organism	<i>Pseudomonas putida</i> KT2440 (PSEPK)	☐ Correct strain
Family	CDP-alcohol phosphatidyltransferase class-II	☐ InterPro IPR016270 (PGS1)
Domains	PLD-like, PLipase_D/transphosphatidylase, PLDc_2 (PF13091)	☐ Confirmed by InterPro/Pfam scan of Q88GQ4

The gene symbol *pssA* is unambiguous in this context and aligns fully with the protein description, organism, family, and domain architecture. A nomenclature caveat: in *E. coli* and *P. putida*, "PssA" denotes a **PLD-superfamily** phosphatidylserine synthase (PSS type II), whereas some organisms (e.g., yeast) use a mechanistically distinct **base-exchange** type I PSS. The *P. putida* enzyme is unambiguously the **type II / PLD-superfamily** enzyme, as established by its HKD motifs and PLD-domain signatures.

Key Findings

Finding 1 — PssA is phosphatidylserine synthase, the committed step of PE biosynthesis

The core function of PP_3664 is synthesis of phosphatidylserine as the obligatory precursor of phosphatidylethanolamine. Curated annotation assigns Q88GQ4 the activity **CDP-diacylglycerol—serine O-phosphatidyltransferase (EC 2.7.8.8)**, corresponding to **GO:0003882**. The reaction catalyzed is:



Phosphatidylserine is subsequently decarboxylated by phosphatidylserine decarboxylase (Psd) to phosphatidylethanolamine (PE). This makes PssA the **committed, rate-influencing entry enzyme** of the PE branch of glycerophospholipid metabolism.

The functional assignment is supported by genetic evidence from multiple orthologs. A recent structural study describes PssA plainly as "*essential in the biosynthesis of phosphatidylethanolamine, a major phospholipid of bacterial membranes*" (PMID: 39693441). In *E. coli*, deletion of most of the *pssA* gene produced a mutant that "*lacked phosphatidylserine synthase and phosphatidylethanolamine and required divalent metal ions for growth*" (PMID: 8824831) — demonstrating that *pssA* is required both for the enzymatic activity and for the downstream lipid product. The requirement is conserved across Gram-negative bacteria: in *Sinorhizobium meliloti*, "*inactivation of the sinorhizobial pssA gene leads to the inability to form PE*" (PMID: 14996797). Together these establish PssA as the necessary and committed producer of PS en route to PE.

Finding 2 — PssA is a PLD-superfamily enzyme using a covalent intermediate, and is a peripheral membrane protein

A direct domain and sequence analysis of Q88GQ4 places PssA firmly in the **phospholipase D (PLD) superfamily**. The InterPro/Pfam signatures are **IPR001736** (Phospholipase D/transphosphatidylase), **IPR025202** (PLD-like domain), **PF13091** (PLDc_2), **SUPFAM SSF56024** (Phospholipase D/nuclease), and **IPR016270** (PGS1). A scan of the 442-amino-acid sequence reveals the **two canonical PLD-superfamily HKD catalytic motifs** (H-x-K-x₄-D):

Motif	Sequence	Position (approx.)
HKD motif I	HLKGSIID	residue ~130
HKD motif II	HAKGVWID	residue ~348

These two half-site motifs juxtapose in the folded protein to form a single bilobed active site — a hallmark of the PLD superfamily, in which catalysis proceeds via a **covalent phosphatidyl-histidine intermediate** rather than direct in-line transfer.

The catalytic mechanism and localization are described in detail for the *E. coli* ortholog: "*A peripheral membrane protein PssA can associate with the cellular membrane in its active state or exist in the cytosol in an inactive form. The membrane-bound enzyme acts on cytidine diphosphate diacylglycerol (CDP-DG) to form cytidine monophosphate and a covalent intermediate, which is subsequently targeted by serine to produce phosphatidylserine*" (PMID: 39693441). This two-step, ping-pong-like mechanism — (1) the enzyme attacks CDP-DG, releasing CMP and forming a phosphatidyl-enzyme covalent intermediate; (2) serine attacks the intermediate to yield PS — is fully consistent with the two HKD motifs found in Q88GQ4. UniProt further annotates the enzyme with **GO:0005829 (cytosol)**, reflecting the fact that PssA is a **peripheral** rather than integral membrane protein: it partitions between an inactive cytosolic pool and an active membrane-associated pool.

Finding 3 — PE, the pathway's downstream product, is a dominant *P. putida* membrane lipid and a stress-adaptation lever

The physiological relevance of the PssA pathway in *P. putida* is underscored by lipidomic analysis. High-resolution LC/MS lipidomics of *P. putida* KT2440 identified "305 distinct glycerophospholipids [e.g. PE(16:0/16:1)]" (PMID: 21895997), with **phosphatidylethanolamine among the principal head-group classes**. Because PssA supplies the PS that is obligately decarboxylated to PE, the enzyme sits directly upstream of this dominant lipid pool. Membrane head-group composition is remodeled under solvent (n-butanol) stress, indicating that flux through the PE branch is physiologically dynamic.

Direct engineering evidence links PssA level to membrane robustness. In a metabolic-engineering study, "*increasing the expression of phosphatidylserine synthase (+pssA) was found to significantly increase both the tolerance and production of octanoic acid*" (PMID: 28867349). This demonstrates that PssA expression is a control point governing PE abundance and, through it, membrane integrity and tolerance to membrane-damaging compounds — a property of particular importance in the solvent-tolerant bacterium *P. putida*.

Finding 4 — *P. putida* PssA is a confident ortholog of the crystallized *E. coli* PssA (46% identity)

To justify transferring the *E. coli* structural/mechanistic model to the *P. putida* protein, a global Needleman–Wunsch alignment was performed between Q88GQ4 (442 aa) and *E. coli* PssA (P23830, 451 aa). The alignment gave **204/440 = 46.4% identity** over the aligned length — well above the ~30% "twilight-zone" threshold for confident orthology, indicating clear one-to-one functional equivalence. Critically, both proteins carry the two HKD catalytic motifs in the same register:

	HKD motif I	HKD motif II
<i>E. coli</i> PssA (P23830)	HFKGFIID (res ~138)	HLKGMWVD (res ~357)
<i>P. putida</i> PssA (Q88GQ4)	HLKGSIID (res ~130)	HAKGVWID (res ~348)

The 2024 *E. coli* PssA crystal structures resolved both an apo form and a CDP-DG-bound Michaelis complex: "*Here, we present two crystal structures of Escherichia coli PssA, one complexed with CDP-DG and the other without. The lipid-bound structure mimics the Michaelis complex before the formation of a covalent intermediate, revealing key determinants for substrate recognition and catalysis*" (PMID: 39693441). The structures also revealed a regulatory mechanism: "*membrane-free PssA is in a monomer-dimer equilibrium, with only the monomer capable of associating with the membrane, suggesting a regulatory mechanism for phospholipid biosynthesis dependent on the oligomerization state of the enzyme*" (PMID: 39693441). Given 46% identity and conserved catalysis, these substrate-recognition, catalytic, and oligomerization-based regulatory insights transfer with high confidence to the *P. putida* enzyme.

Finding 5 — PssA (PP_3664) is the entry enzyme of a two-step PE module; Psd is separately encoded at PP_4908

KEGG assigns **ppu:PP_3664** to orthology **K00998** (CDP-diacylglycerol—serine O-phosphatidyltransferase, EC 2.7.8.8) and to module **ppu_M00093**, "*Phosphatidylethanolamine (PE) biosynthesis, PA => CDP-DAG => PS => PE,*" with gene coordinates complement(4162133..4163461). The downstream phosphatidylserine decarboxylase (K01613, Psd) maps to a **distinct, non-adjacent locus, ppu:PP_4908**. Thus, unlike the *pss-psd* operon of *Bacillus subtilis*, the *P. putida* *pssA* and *psd* genes are **not genomically clustered**, and the pathway is assembled from two separately regulated loci.

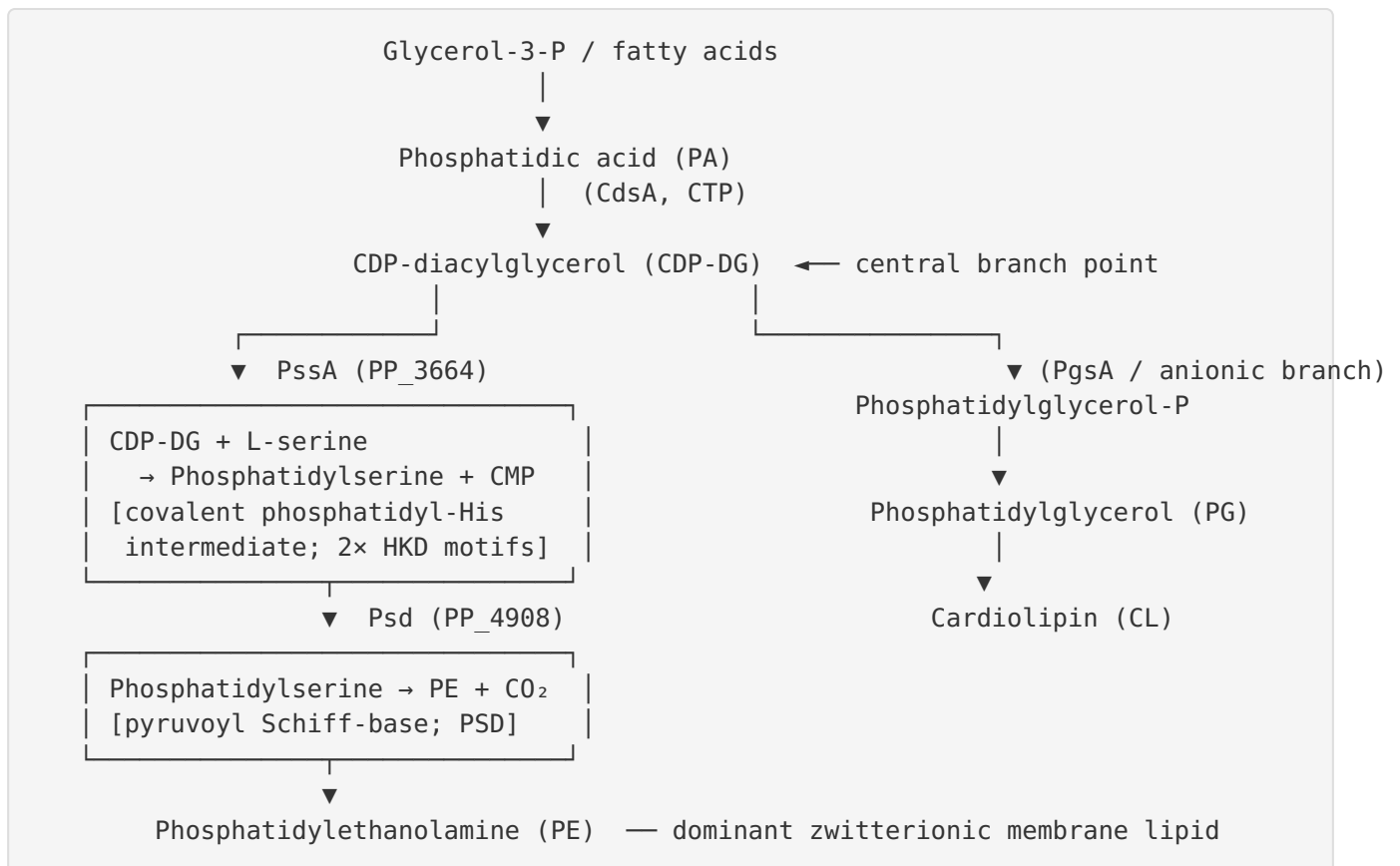
The chemistry of the downstream step is well defined by structural studies of *E. coli* PSD: "Phosphatidylethanolamine (PE), a major component of the cellular membrane across all domains of life, is synthesized exclusively by membrane-anchored phosphatidylserine decarboxylase (PSD) in most bacteria. The enzyme undergoes auto-cleavage for activation and utilizes the pyruvoyl moiety to form a Schiff base intermediate with PS to facilitate decarboxylation" (PMID: 33707636). A companion structural study of *E. coli* Psd reported that "EcPsd forms a homodimer, and each protomer has a positively charged substrate binding pocket at the active site... EcPsd has an N-terminal hydrophobic helical region that is important for membrane binding" (PMID: 32402247). PssA thus produces the substrate (PS) that the pyruvoyl-dependent, membrane-anchored Psd converts to PE, completing the module. Because PS does not normally accumulate, PssA activity effectively sets the supply of PE.

Finding 6 — The serine-binding (acceptor) site is somewhat promiscuous

The head-group acceptor site of PssA is not strictly limited to L-serine. In *E. coli*, an unidentified membrane lipid observed in minimal medium was characterized by LC-MS and metabolic labeling as **phosphatidylhomoserine (PHS)**, and it was shown to be produced by PssA: "In *E. coli*, PHS was produced endogenously by phosphatidylserine synthase A (PssA), confirmed by the absence of PHS in an *E. coli* $\Delta pssA$ mutant" (PMID: 40404014). This indicates the acceptor site can accommodate **L-homoserine** in addition to its physiological substrate L-serine — measurable substrate promiscuity. Given the 46% identity to *E. coli* PssA, the *P. putida* enzyme plausibly shares this capacity, though it has not been tested directly. This informs, but does not change, the physiological specificity for L-serine.

Mechanistic Model / Interpretation

Integrating the findings yields a coherent picture of PssA's role in *P. putida* membrane biogenesis:



Location of function. PssA acts at the **cytoplasmic (inner) leaflet of the inner membrane**, where CDP-DG resides, but is itself a **peripheral membrane protein** that shuttles between an inactive cytosolic pool and an active membrane-bound state. Membrane association is coupled to its oligomerization state — the monomer binds membrane and is catalytically productive, while the dimer partitions away — providing a built-in regulatory switch that couples enzyme activity to the availability of its lipid substrate and to the physical state of the membrane.

Pathway role and branch-point control. CDP-DG is the shared precursor for both the **zwitterionic branch** (PS → PE, via PssA and Psd) and the **acidic branch** (phosphatidylglycerol and cardiolipin, via PgsA/Cls). By committing CDP-DG to serine, PssA governs the partitioning of flux between these two branches and therefore helps set the **PE : PG/CL ratio** — a determinant of membrane surface charge, curvature stress, and bilayer integrity. This explains why manipulating PssA levels alters solvent/organic-acid tolerance ([PMID: 28867349](#)): more PssA drives more PE, remodeling the membrane's biophysical properties.

Catalytic logic. As a PLD-superfamily transferase, PssA does not perform a direct one-step transfer. Instead its two HKD half-sites cooperate: one histidine attacks the phosphodiester of CDP-DG to expel CMP and form a **covalent phosphatidyl-enzyme intermediate**; the second histidine positions L-serine's hydroxyl for nucleophilic attack on the intermediate, releasing

PS. This transphosphatidylation chemistry — the same mechanistic class as phospholipase D — accounts for the enzyme's ability to occasionally accept alternative alcohols such as L-homoserine (Finding 6).

Evidence Base

PMID	Title (abbreviated)	Relevance to PssA (PP_3664)
39693441	<i>Structural basis for membrane association and catalysis by phosphatidylserine synthase</i>	Primary structural/mechanistic source. Crystal structures of <i>E. coli</i> PssA (apo + CDP-DG complex); defines peripheral-membrane localization, covalent-intermediate catalysis, and monomer–dimer regulation. Transfers to Q88GQ4 via 46% identity.
8824831	<i>A regulatory mechanism for the balanced synthesis of membrane phospholipid species in E. coli</i>	Genetic knockout: <i>E. coli</i> $\Delta pssA$ lacks PS synthase activity and PE; establishes necessity of <i>pssA</i> for the pathway.
14996797	<i>Phosphatidylethanolamine is not essential for growth of Sinorhizobium meliloti...</i>	Independent Gram-negative ortholog: <i>pssA</i> inactivation abolishes PE formation.
21895997	<i>The glycerophospholipid inventory of Pseudomonas putida...</i>	Organism-specific lipidomics: PE is a principal glycerophospholipid class in <i>P. putida</i> KT2440 — the downstream product of the PssA route.
28867349	<i>Engineering E. coli membrane phospholipid head distribution improves tolerance...</i>	Overexpressing <i>pssA</i> raises PE and boosts tolerance/production of a fatty acid — links PssA level to membrane robustness.
33707636	<i>Structural insights into phosphatidylethanolamine formation...</i>	Defines the downstream Psd step (pyruvoyl Schiff-base decarboxylation of PS to PE).
32402247	<i>Structural Basis for PE Biosynthesis by Bacterial Phosphatidylserine Decarboxylase</i>	<i>E. coli</i> Psd crystal structures; membrane-binding N-terminal helix and active-site determinants of the enzyme acting on PssA's product.
40404014	<i>Endogenous formation of phosphatidylhomoserine in E. coli through PssA</i>	Demonstrates acceptor-site promiscuity: PssA can use L-homoserine to make phosphatidylhomoserine.
9422599	<i>Cloning... of the B. subtilis psd gene</i>	Context: in <i>B. subtilis</i> , <i>pss</i> and <i>psd</i> are adjacent (operon), contrasting with the separated <i>P. putida</i> loci.

The evidence base is internally consistent. The strongest direct experimental data (knockouts, crystal structures) come from orthologs — principally *E. coli* — and are transferred to Q88GQ4 on the basis of unambiguous orthology (46% identity, conserved twin HKD motifs, identical domain architecture). Organism-specific support for *P. putida* comes from lipidomics (PE is abundant) and KEGG pathway placement (PP_3664 → K00998 → module M00093).

Supported and Refuted Hypotheses

- **Supported:** PssA is a phosphatidylserine synthase performing the committed step of PE biosynthesis; it is a PLD-superfamily peripheral membrane enzyme using a covalent phosphatidyl-His mechanism; it acts at the inner-membrane cytoplasmic face; it is upstream of a separately encoded Psd (PP_4908); PE is a dominant, physiologically important lipid in *P. putida*.
 - **Refuted / excluded:** PssA is **not** an integral-membrane, base-exchange type-I phosphatidylserine synthase; despite the automated "PGS1"/PGP-synthase InterPro cross-reference, the curated EC assignment, KEGG orthology (K00998), sequence analysis, and genomic/physiological context identify the enzyme as a **serine**-specific phosphatidyltransferase (PS synthase), not a phosphatidylglycerophosphate synthase. The shared PLD fold among PS synthase, eukaryotic PGP synthase, and cardiolipin synthase is a family-level artifact of the annotation, not evidence of PGP-synthase activity.
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Limitations and Knowledge Gaps

1. **No direct biochemical characterization of the *P. putida* KT2440 protein.** All enzymatic, structural, and mechanistic details are inferred from orthologs (chiefly *E. coli*) via strong sequence homology. The kinetic parameters, exact substrate specificity, and regulation of the actual PP_3664 gene product have not been measured directly.
2. **No *P. putida* pssA knockout phenotype in the literature reviewed.** While *E. coli* and *S. meliloti* pssA deletions abolish PE, the specific consequences of deleting PP_3664 in *P. putida* (essentiality, divalent-cation dependence, solvent-tolerance defects) were not directly documented.

3. **Acceptor-site promiscuity is untested in *P. putida*.** Phosphatidylhomoserine formation was demonstrated in *E. coli*; whether the *P. putida* enzyme shares this promiscuity, and whether PHS or other minor head-group variants occur in *P. putida* membranes, remains unknown.
4. **Regulation of PssA membrane association in *P. putida*.** The monomer–dimer/membrane-association switch was established for *E. coli*; the specific determinants (lipid composition, ionic conditions, protein partners) governing this in *P. putida* have not been examined.
5. **Sequence-motif positions are approximate.** The HKD motif residue numbers and identity percentage derive from in-silico alignment; while robust, they have not been experimentally validated by mutagenesis in this organism.
6. **Divalent-cation / Mg^{2+} requirement not confirmed for PP_3664.** *E. coli* PssA-null mutants require divalent metal ions for growth; whether *P. putida* PssA activity or the mutant phenotype depends similarly on Mg^{2+}/Ca^{2+} was not directly established.

Proposed Follow-up Experiments / Actions

1. **Targeted gene deletion of PP_3664 in *P. putida* KT2440.** Construct an in-frame $\Delta pssA$ mutant and quantify membrane phospholipids by LC-MS. Prediction: loss of PS synthase activity and PE, with possible divalent-cation dependence — directly testing transferability of the *E. coli* phenotype.
2. **In-vitro enzyme assay with purified recombinant PssA (Q88GQ4).** Express and purify the protein; measure PS synthase activity from CDP-DG + L-serine, determine kinetic constants (K_m , k_{cat}), and test the Mg^{2+}/Mn^{2+} requirement.
3. **Substrate-specificity / promiscuity screen.** Offer L-homoserine, L-threonine, and other candidate alcohols as acceptors to test whether the *P. putida* enzyme, like *E. coli* PssA, produces phosphatidylhomoserine or other head-group variants.
4. **Site-directed mutagenesis of the two HKD motifs** (His130, His348 and associated Lys/Asp residues) to confirm their catalytic essentiality and validate the covalent-intermediate mechanism in this ortholog.

5. **Membrane-association / oligomerization studies.** Use sedimentation/flotation assays and native MS or size-exclusion chromatography to test whether *P. putida* PssA exhibits the monomer-active/dimer-inactive regulatory behavior seen in *E. coli*, and identify lipid dependencies.
6. **Structural determination or AlphaFold-model validation.** Solve the crystal/cryo-EM structure of *P. putida* PssA, or validate an AlphaFold model against the *E. coli* templates, to confirm active-site architecture and substrate-recognition determinants.
7. **Stress-tolerance engineering in *P. putida*.** Titrate *pssA* expression and assay solvent/organic-acid tolerance and PE content, extending the *E. coli* engineering result ([PMID: 28867349](#)) to the native solvent-tolerant host — of biotechnological relevance.

Conclusion

The gene *pssA* (Q88GQ4, PP_3664) of *Pseudomonas putida* KT2440 encodes **phosphatidylserine synthase (EC 2.7.8.8)**, a **phospholipase-D-superfamily CDP-diacylglycerol—serine O-phosphatidyltransferase** that catalyzes the committed first step of phosphatidylethanolamine biosynthesis. It transfers the phosphatidyl group of CDP-diacylglycerol onto L-serine — via a covalent phosphatidyl-histidine intermediate built from two conserved HKD motifs — to form phosphatidylserine and CMP; a separate decarboxylase (Psd, PP_4908) then converts PS to PE, the dominant membrane phospholipid of *P. putida*. The enzyme functions as a **peripheral membrane protein at the cytoplasmic face of the inner membrane**, its activity gated by membrane association and oligomerization state, and it governs the CDP-diacylglycerol branch point that balances zwitterionic (PE) versus acidic (PG/cardiolipin) phospholipid synthesis. These conclusions rest on curated EC/KEGG/GO annotation, InterPro/Pfam domain signatures and direct sequence analysis, and strong orthology (46% identity, conserved catalytic motifs) to the experimentally characterized and crystallized *E. coli* PssA.

Prepared through iterative literature review and bioinformatic analysis (UniProt, KEGG, InterPro/Pfam, sequence alignment, and primary structural/genetic literature).

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